

■ Chili Cultivation Guide

A Complete Guide for Growing Chili Peppers at Home

1. Introduction

Chili peppers (*Capsicum annuum* and related species) are one of the easiest and most rewarding crops to grow at home. They thrive in pots, raised beds, or open ground, and reward growers with months of continuous harvest. This guide walks you through every stage — from seed selection to harvesting and pest management.

2. Climate and Site Requirements

- **Temperature:** Chilies grow best between 20–30°C (68–86°F). Growth slows below 15°C and above 35°C.
- **Sunlight:** Requires 6–8 hours of direct sunlight daily.
- **Soil:** Well-drained, loamy soil rich in organic matter, with a pH of 6.0–6.8.
- **Location:** Choose a spot sheltered from strong winds; balconies, terraces, and backyard beds all work well.

3. Choosing Seeds and Varieties

Select varieties suited to your taste and climate. Popular home-garden types include:

- **Bird's Eye Chili** – very hot, compact plant, ideal for pots.
- **Cayenne** – medium-hot, long fruits, prolific yields.
- **Bell/Capsicum** – mild, large fruit, great for salads and cooking.
- **Jalapeño** – medium heat, thick-walled, versatile.
- **Kashmiri/Guntur Chili** – popular Indian varieties, good for drying and powder.

4. Sowing and Germination

- Soak seeds in warm water for 6–8 hours before sowing to speed up germination.
- Sow seeds 0.5 cm deep in a seed tray or small pots filled with a light seed-starting mix.
- Keep soil consistently moist (not waterlogged) and maintain a temperature of 25–28°C.
- Germination typically takes 7–14 days.
- Once seedlings develop 4–6 true leaves (about 4–6 weeks), they are ready for transplanting.

5. Transplanting

- Harden off seedlings for 3–5 days by gradually exposing them to outdoor conditions.
- Transplant into pots (minimum 30 cm diameter, 30 cm depth) or garden beds spaced 45–60 cm apart.
- Plant in the early morning or evening to reduce transplant shock.
- Water thoroughly immediately after transplanting.

6. Watering

Chili plants prefer evenly moist soil but are sensitive to overwatering, which can cause root rot. Water deeply 2–3 times a week, adjusting for rainfall and temperature. Reduce watering slightly once fruiting begins to intensify flavor and heat. Mulching around the base helps retain moisture and suppress weeds.

7. Fertilizing and Nutrition

- **Base dose:** Mix well-rotted compost or farmyard manure into the soil before planting.
- **Vegetative stage:** Apply a balanced NPK fertilizer (e.g., 19:19:19) every 3–4 weeks.
- **Flowering/fruiting stage:** Switch to a fertilizer higher in potassium and phosphorus (e.g., 13:0:45 or bone meal) to boost flowering and fruit set.
- Add calcium (crushed eggshells or gypsum) to prevent blossom end rot.

8. Support and Pruning

- Stake tall or heavily fruiting varieties to prevent branches from breaking.
- Pinch off the first few flower buds to encourage stronger vegetative growth before fruiting.
- Prune lower leaves that touch the soil to improve airflow and reduce disease risk.

9. Common Pests and Diseases

Problem	Symptoms	Organic Control
Aphids	Curled leaves, sticky residue	Neem oil spray, ladybugs
Whiteflies	Yellowing leaves, tiny white flies	Yellow sticky traps, neem oil
Fruit Borer	Holes in fruit, larvae inside	Remove affected fruit, neem spray
Powdery Mildew	White powder on leaves	Improve airflow, baking soda spray
Blossom End Rot	Dark, sunken spot on fruit base	Consistent watering, add calcium
Damping Off	Seedlings collapse at soil line	Avoid overwatering, use sterile mix

10. Flowering, Fruiting and Harvest

- Flowering usually begins 6–8 weeks after transplanting.
- Fruits typically mature 20–30 days after flowering, changing from green to red, yellow, or orange depending on variety.
- Harvest by cutting (not pulling) the stem just above the fruit to avoid damaging branches.
- Regular harvesting encourages the plant to keep producing new flowers and fruit.

11. Drying and Storage

- **Sun drying:** Spread whole chilies on a mesh tray in full sun for 7–10 days until brittle.
- **Oven drying:** Dry at the lowest oven setting (around 60°C) with the door slightly open for 6–8 hours.
- **Storage:** Store dried chilies or powder in an airtight container away from light and moisture.
- Fresh chilies can be refrigerated for 1–2 weeks or frozen for several months.

12. Quick Care Calendar

Week	Stage	Key Task
0	Sowing	Soak & sow seeds indoors
2	Germination	Keep warm & moist
4–6	Seedling	Harden off
6–7	Transplant	Move to pot/bed
8–10	Vegetative	Balanced fertilizer

10–12	Flowering	Switch to P–K fertilizer
12+	Fruiting/Harvest	Harvest regularly

13. Pro Tips

- Rotate chili crops each season to reduce soil-borne disease build-up.
- Grow marigolds nearby — they help repel common pests.
- Pick a few peppers even before you need them; frequent picking boosts overall yield.
- In containers, feed more often since nutrients leach out faster with watering.

Happy Growing! ■